4.05 Further Algebra
Relationships between roots of and coefficients of polynomials are explored. Techniques involving partial fractions are developed.

software to investigate the relationships between graphical and algebraic representations, including complex numbers, vectors, implicit and parametric curves, and differential equations.

2. Computer Algebra System (CAS): Learners could use CAS software to investigate algebraic relationships, including derivatives and integrals, to solve equations, including differential equations, to manipulate matrices and as an investigative problem solving tool. This is best done in conjunction with other software such as graphing tools and spreadsheets.

3. Visualisation: Learners could use appropriate software to visualise situations in 3-D relating to lines and planes, and to linear transformations.

4. Spreadsheets: Learners could use spreadsheet software to investigate numerical methods, sequences and series, for modelling and to generate tables of values for functions.

Content of Pure Core (Mandatory papers Y540 and Y541)

When this course is being co-taught with OCR's AS Level Further Mathematics A (H235) the 'Stage 1' column indicates the common content between the two specificatio and the 'Stage 2' column indicates content which is particular to this specification. In each section the OCR Reference lists 'Stage 1' statements before 'Stage 2' statement

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.01 Proof				
4.01a 4.01b	Mathematical induction	a) Be able to construct proofs using mathematical induction. *This topic may be tested using any relevant content including divisibility, powers of matrices and results on powers, exponentials and factorials.* *e.g. Prove that* $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix}$ *for* $n \in \mathbb{Z}^+$. *Prove that* $7^n - 3^n$ *is divisible by* 4 *for* $n \in \mathbb{Z}^+$. *Prove that* $2^n > 2n$ *for* $n \geq 3, n \in \mathbb{Z}$.	b) Be able to construct proofs of a more demanding nature, including conjecture followed by proof. *This topic may be tested using any relevant content including sums of series.* *e.g. Prove that* $\sum_{r=1}^{n} \frac{1}{r(r+1)} = \frac{n}{n+1}$ *for all* $n \in \mathbb{Z}$. *Prove that* $3^n > n^3$ *for* $n \geq 4, n \in \mathbb{Z}$. *Prove that* $n! < n^n$ *for* $n \geq 1, n \in \mathbb{Z}$. *Prove that* $(1+x)^n \geq 1 + nx$ *for any real number* $x > -1$ *and* $n \in \mathbb{Z}^+$. *Prove that the* n*th derivative of* $x^2 e^x$ *is* $(x^2 + 2nx + n(n-1))e^x$.	A1

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4.02 Complex Numbers				
4.02a	The language of complex numbers	a) Understand the language of complex numbers. *Know the meaning of "real part", "imaginary part", "conjugate", "modulus" and "argument" of a complex number.*		B2 B5 B3
4.02b 4.02d		b) Be able to express a complex number z in either cartesian form $z = x + \mathrm{i}y$, where $\mathrm{i}^2 = -1$, or modulus-argument form $z = r(\cos\theta + \mathrm{i}\sin\theta) = [r, \theta] = r\mathrm{cis}\theta$, where $r \geq 0$ is the modulus of z and θ, measured in radians is the argument of z.	d) Understand and be able to use the exponential form, $r\mathrm{e}^{\mathrm{i}\theta}$, of a complex number.	B9
4.02c		c) Understand and be able to use the notation: z, z^*, $\mathrm{Re}(z)$, $\mathrm{Im}(z)$, $\arg(z)$, $\lvert z \rvert$. *Includes knowing that a complex number is zero if and only if both the real and imaginary parts are zero.* *The principal argument of a complex number, for uniqueness, will be taken to lie in either of the intervals* $[0,\ 2\pi)$ *or* $(-\pi, \pi]$*. Learners may use either as appropriate unless the interval is specified.* *In stage 1 knowledge of radians is assumed: see H240 section 1.05d.*		

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.02e 4.02f	Basic operations	e) Be able to carry out basic arithmetic operations $(+,-,\times,\div)$ on complex numbers in both cartesian and modulus-argument forms. *In stage 1 knowledge of radians and compound angle formulae is assumed: see H240 sections 1.05d and 1.05l.* *Learners may use the results* $z_1z_2=[r_1r_2,\theta_1+\theta_2]$ *and* $\frac{z_1}{z_2}=\left[\frac{r_1}{r_2},\theta_1-\theta_2\right]$. f) Convert between cartesian and modulus-argument forms.		B2 B5 B6
4.02g 4.02h 4.02i 4.02j	Solution of equations	g) Know that, for a polynomial equation with real coefficients, complex roots occur in conjugate pairs. h) Be able to find algebraically the two square roots of a complex number. *e.g. By squaring and comparing real and imaginary parts.* i) Be able to solve quadratic equations with real coefficients and complex roots. j) Be able to use conjugate pairs, and the factor theorem, to solve or factorise cubic or quartic equations with real coefficients. *Where necessary, sufficient information will be given to deduce at least one root for cubics or at least one complex root or quadratic factor for quartics.*		B1 B3

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.02k 4.02m 4.02l	Argand diagrams	k) Be able to use and interpret Argand diagrams. *e.g. To represent and interpret complex numbers geometrically.* *Understand and use the terms "real axis" and "imaginary axis".* l) Understand the geometrical effects of taking the conjugate of a complex number, and adding and subtracting two complex numbers.	m) Understand the geometrical effects of multiplying and dividing two complex numbers. *Includes raising complex numbers to positive integer powers.*	B4 B6
4.02n	Euler's formula		n) Know and be able to use Euler's formula $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$. *e.g. To express, and work with, complex numbers in the forms* $r(\cos\theta + \mathrm{i}\sin\theta) = r\mathrm{e}^{\mathrm{i}\theta} = r\mathrm{cis}\,\theta = [r,\theta]$.	B6 B9

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.02o 4.02p	Loci	o) Be able to illustrate equations and inequalities involving complex numbers by means of loci in an Argand diagram. *i.e. Circle of the form* $\lvert z-a\rvert = k$, *half-lines of the form* $\arg(z-a) = b$, *lines of the form* $\mathrm{Re}(z) = k$, $\mathrm{Im}(z) = k$ *and* $\lvert z-a\rvert = \lvert z-b\rvert$, *and regions defined by inequalities in these forms.* *To include the convention of dashed and solid lines to show exclusion and inclusion respectively.* *No shading convention will be assumed. If not directed, learners should indicate clearly which regions are included.* *In stage 1 knowledge of radians is assumed: see H240 section 1.05d.* p) Understand and be able to use set notation in the context of loci. *e.g. The region* $\lvert z-a\rvert > k$ *where* $z = x + \mathrm{i}y$, $a = x_a + \mathrm{i}y_a$ *and* $k > 0$ *may be represented by the set* $\{x+\mathrm{i}y : (x-x_a)^2 + (y-y_a)^2 > k^2\}$. *In stage 1 knowledge of radians is assumed: see H240 section 1.05d.*		B7

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
4.02q	De Moivre's theorem		q) Understand de Moivre's theorem and use it to find multiple-angle formulae and sums of series involvin trigonometric and/or exponential terms. *Express trigonometrical ratios of multiple angles in terms c powers of trigonometrical ratios of the fundamental angle* *e.g.* $\sin(3\theta) = 3\sin\theta - 4\sin^3\theta$. *Use expressions for* $\sin\theta$ *and* $\cos\theta$ *in terms of* $e^{i\theta}$ *or equivalent relationships.* *e.g.* $\sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2i}$. *Express powers of* $\sin\theta$ *and* $\cos\theta$ *in terms of series of trigonometric ratios of multiples of the fundamental angle* *e.g.* $\sin^5\theta = \frac{1}{16}(10\sin\theta - 5\sin(3\theta) + \sin(5\theta))$.
4.02r	nth roots		r) Be able to find the n distinct nth roots of $re^{i\theta}$ for $r \neq$ and know that they form the vertices of a regular n-gon on an Argand diagram. *Answers may be asked for in either cartesian or modulus-argument form.*
4.02s	Roots of unity		s) Be able to use complex roots of unity to solve geometric problems. *e.g. To locate the roots of unity on an Argand diagram or t prove results about sums of roots of unity.*

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.03 Matrices				
4.03a	The language of matrices	a) Understand the language of matrices. *Understand the meaning of "conformable", "equal", "square", "rectangular", "m by n", "determinant", "zero" and "null", "transpose" and "identity" when applied to matrices.* *Learners should be familiar with real matrices and complex matrices.*		C2
4.03b	Matrix addition and multiplication	b) Be able to add, subtract and multiply conformable matrices; multiply a matrix by a scalar. *Learners may perform any operations involving entirely numerical matrices by calculator.* *Includes raising square matrices to positive integer powers.* *Learners should understand the effects on a matrix of adding the zero matrix to it, multiplying it by the zero matrix and multiplying it by the identity matrix.*		C1
4.03c		c) Understand that matrix multiplication is associative but not commutative. *Understand the terms "associative" and "commutative".*		

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4.03d	Linear transformations	d) Be able to find and use matrices to represent linear transformations in 2-D. *Includes:* • *reflection in either coordinate axis and in the lines* $y = \pm x$ • *rotation about the origin (defined by the angle of rotation* θ*, where the direction of positive rotation is taken to be anticlockwise)* • *enlargement centre the origin (defined by the the scale factor)* • *stretch parallel to either coordinate axis (defined by the invariant axis and scale factor)* • *shear parallel to either coordinate axis (defined by the invariant axis and the image of a transformed point).* *Includes the terms "object" and "image".*		C3
4.03e		e) Be able to find and use matrices to represent successive transformations. *Includes understanding and being able to use the result that the matrix product* $\mathbf{AB}$ *represents the transformation that results from the transformation represented by* $\mathbf{B}$ *followed by the transformation represented by* $\mathbf{A}$.		

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.03f		f) Be able to use matrices to represent single linear transformations in 3-D. *3-D transformations will be confined to reflection in one of the planes* $x = 0, y = 0, z = 0$ *or rotation about one of the coordinate axes. The direction of positive rotation is taken to be anticlockwise when looking towards the origin from the positive side of the axis of rotation.* *Includes the terms "plane of reflection" and "axis of rotation".* *In stage 1 knowledge of 3-D vectors is assumed: see H240 section 1.10b.*		C3
4.03g	Invariance	g) Be able to find invariant points and lines for a linear transformation. *Includes the distinction between invariant lines and lines of invariant points.* [*The 3-D transformations in section 4.03f are excluded.*]		C4
4.03h 4.03i	Determinants	h) Be able to find the determinant of a 2×2 matrix with and without a calculator. *Use and understand the notation* $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ *or* $\lvert \mathbf{M} \rvert$ *or* $\det \mathbf{M}$. i) Know that the determinant of a 2×2 matrix is the area scale factor of the transformation defined by that matrix, including the effect on the orientation of the image. *Learners should know that a transformation preserves the orientation of the object if the determinant of the matrix which represents it is positive and that the transformation reverses orientation if the determinant*		C5

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.03j		j) Be able to calculate the determinant of a 3×3 matrix with and without a calculator.		C5
4.03k		k) Know that the determinant of a 3×3 matrix is the volume scale factor of the transformation defined by that matrix, including the effect on the preservation of the orientation of the image. *Learners should know that the sign of the determinant determines whether or not the corresponding transformation preserves orientation, but do not need to understand the geometric interpretation of this in 3-D.*		
4.03l		l) Understand and be able to use singular and non-singular matrices. *Includes understanding the significance of a zero determinant.*		
4.03m		m) Know and be able to use the result that $\det(\mathbf{AB}) = \det(\mathbf{A}) \times \det(\mathbf{B})$.		
4.03n	Inverses	n) Be able to find and use the inverse of a non-singular 2×2 matrix with and without a calculator.		C6
4.03o		o) Be able to find and use the inverse of a non-singular 3×3 matrix with and without a calculator.		
4.03p		p) Understand and be able to use simple properties of inverse matrices. *e.g. The result that* $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.		
4.03q		q) Understand and be able to use the connection between inverse matrices and inverse transformations.		

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.03r 4.03s	Solution of simultaneous equations	r) Be able to solve two or three linear simultaneous equations in two or three variables by the use of an inverse matrix, where a unique solution exists.	s) Be able to determine, for two or three linear simultaneous equations where no unique solution exists, whether the equations have an infinite set of solutions (the equations are consistent) or no solutions (the equations are inconsistent). [*Finding the solution set in the infinite case is excluded.*]	C7
4.03t	Intersection of planes		t) Be able to interpret the solution or failure of solution of three simultaneous linear equations in terms of the geometrical arrangement of three planes. *Learners should know and be able to identify the different ways in which two or three distinct planes can intersect in 3-D space, including cases where two or three of the planes are parallel.* *Learners should understand and be able to apply the geometric significance of the singularity of a matrix in relation to the solution(s) or non-existence of them.* [*Finding the line of intersection of two or more planes is excluded.*]	C8

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.04 Further Vectors				
4.04a	Equation of a straight line	a) Understand and be able to use the equation of a straight line, in 2-D and 3-D, in cartesian and vector form. *Learners should know and be able to use the forms:* $y = mx + c$, $ax + by = c$ *and* $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ *in 2-D, and* $\frac{x - a_1}{u_1} = \frac{y - a_2}{u_2} = \frac{z - a_3}{u_3} (= \lambda)$ *and* $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ *in 3-D.* *Includes being able to convert from one form to another.*		F1
4.04b	Equation of a plane		b) Understand and be able to use the equation of a plane in cartesian and vector form. *Learners should know and be able to use the forms:* $ax + by + cz = d$, $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$, $(\mathbf{r} - \mathbf{a}).\mathbf{n} = 0$ *and* $\mathbf{r}.\mathbf{n} = p$. *Includes being able to convert from one form to another.*	F2
4.04c 4.04d	Scalar product	c) Be able to calculate the scalar product and use it both to calculate the angles between vectors and/or lines, and also as a test for perpendicularity. *Includes the notation* $\mathbf{a.b}$	d) Be able to find the angle between two planes and the angle between a line and a plane.	F3 F4
4.04e 4.04f	Intersections	e) Be able to find, where it exists, the point of intersection between two lines. *Includes determining whether or not lines intersect, are parallel or are skew.*	f) Be able to find the intersection of a line and a plane.	F5

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.04g	Vector product	g) Be able to use the vector product to find a vector perpendicular to two given vectors. *Includes the notation* $\mathbf{a} \times \mathbf{b}$. *When the vector product is required, either a calculator or a formula may be used. The formula below will be given:* $\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & a_1 & b_1 \\ \mathbf{j} & a_2 & b_2 \\ \mathbf{k} & a_3 & b_3 \end{vmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$. [*The magnitude of the vector product is excluded.*]		Essential content for F2, F3, F5, C6
4.04h 4.04i	Shortest distances		h) Be able to find the distance between two parallel lines and the shortest distance between two skew lines. *For skew lines, the formula* $D = \frac{\lvert(\mathbf{b} - \mathbf{a}).\mathbf{n}\rvert}{\lvert\mathbf{n}\rvert}$*, where* $\mathbf{a}$ *and* $\mathbf{b}$ *are position vectors of points on each line and* $\mathbf{n}$ *is a mutual perpendicular to both lines, will be given.* *Either* $\mathbf{n}$ *will be given, or it must be established from given information including by use of the vector product.* i) Be able to find the shortest distance between a point and a line. *The formula* $D = \frac{\lvert ax_1 + by_1 - c\rvert}{\sqrt{a^2 + b^2}}$ *where the coordinates of the point are* (x_1, y_1) *and the equation of the line is given by* $ax + by = c$*, will be given.*	F5

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
4.04j			j) Be able to find the shortest distance between a poir and a plane. *The formula* $D = \frac{\lvert \mathbf{b.n} - p \rvert}{\lvert \mathbf{n} \rvert}$ *where* $\mathbf{b}$ *is the position vector c the point and the equation of the plane is given by* $\mathbf{r.n} = p$, *will be given.*
4.05 Further Algebra			
4.05a	Roots of equations	a) Understand and be able to use the relationships between the symmetric functions of the roots of polynomial equations and the coefficients. *Up to, and including, quartic equations.* *e.g. For the quadratic equation* $ax^2 + bx + c = 0$ *with roots* α *and* β, $\alpha + \beta = -\frac{b}{a}$ *and* $\alpha\beta = \frac{c}{a}$.	
4.05b	Transformation of equations	b) Be able to use a substitution to obtain an equation whose roots are related to those of the original equation. *Equations will be of at least cubic degree.*	
4.05c	Partial fractions		c) Extend their knowledge of partial fractions up to rational functions in which the denominator may include quadratic factors of the form $ax^2 + c$ for $c >$ and in which the degree of the numerator may be equal to, or exceed, the degree of the denominator. *See H240 section 1.02y.*

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